LIVE · UPDATED 23 JUN 2026

METHODOLOGY & THE FULL REPORT

How this was built

Every number on the main page comes from one committed dataset and the code in this repository. Here is what that dataset is, how a stoppage becomes a momentum number, what the comparisons control for, and where the honest limits are.

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00 — FINDINGS IN BRIEF

The team on top of momentum loses about −**25** momentum points in the five minutes after a mandatory hydration break. But the *same* teams lose about −**17** at quiet, break-free minutes — so most of the drop is regression to the mean, the natural cool-off after a hot spell, not the whistle.

What is left is a gap of about **8** points between a break and no break for the same teams. That is consistent with a small real effect, but the intervals overlap and the sample is still small, so it is suggestive, not proven. A separate look at the weather suggests most matches never reached the heat a cooling break is designed for. None of this is a causal verdict yet — the knockouts will sharpen it.

01 — WHAT THIS MEASURES

The question is narrow and testable: do FIFA's mandatory in-match hydration breaks shift momentum away from the team that was on top? The outcome is **FotMob's** per-minute momentum index — their expected-threat model of which side is on top, built from the flow of attacks and chances, not the scoreline. We *read* that number; we do not compute our own. Positive means the home side is pressing, negative the away side.

It is a living analysis: the page rebuilds from the committed dataset every matchday through the July 19 final, so the figures move as data accrues.

02 — WHERE THE DATA COMES FROM

Three sources, each doing one job:

- **FotMob** — the per-minute momentum series, and match lineups (used for the heat check).
- **ESPN** — text commentary, which gives the exact timing of stoppages (and VAR / injury events), so breaks are detected from what actually happened, not a hardcoded clock.
- **Open-Meteo** — temperature and humidity at each venue and kickoff for WBGT, and the historical climate of clubs' home cities for the acclimatization check.

We publish *derived* data only: the processed dataset and dated summaries are committed to the repo, but raw scraped payloads are never redistributed, out of respect for the sources' terms. Scraping runs locally on a home connection; the public site is rebuilt by CI from the committed dataset and never scrapes. Git history is the snapshot system — each day's estimate is a commit.

03 — FROM A STOPPAGE TO A NUMBER

For every detected stoppage we take two five-minute windows of momentum: the **pre** window (the five minutes before the whistle) and the **post** window (the five minutes after), excluding the stoppage minute itself. The outcome is the change between them — `momentum_delta = post mean − pre mean`.

Momentum is home-positive, so each stoppage produces *two* rows, one from each team's perspective (the away row is just the negative). Because they mirror each other, the pooled average is zero by construction. So we always report the team that was **on top** before the break — the "momentum killer" claim is specifically that a break pushes momentum away from whoever was ahead.

04 — THE COMPARISON BASELINES

A team that just had a blazing five minutes tends to cool off anyway, break or no break. That regression to the mean is the single biggest threat to reading the raw drop as an effect. So we run the *exact same measurement* where no break was mandated, on the same FotMob scale, and compare:

- **The same 2026 teams at quiet minutes** (−17) — the cleanest control: identical teams, identical tournament, just no whistle. Anything fixed about a team — its quality, its heat, its altitude — is differenced out here.
- **World Cup 2022** (−15) and **Euro 2024** (−16) — national-team football in cooler settings, no mandated breaks.
- **Copa América 2024** (−10) — national teams in US summer heat, but a noisier single edition.
- **Club World Cup 2025** (−23) — clubs, which regress harder than national teams; a contrast, not a like-for-like.

National-team football regresses about −10 to −17 on its own. The break (−25) sits about 8 points below the same-teams control — the part regression to the mean does not explain.

05 — CONFIDENCE INTERVALS

Several stoppages within one match are not independent, so intervals are bootstrapped by resampling *matches*, not rows (a cluster bootstrap). Early in the tournament there are few match-clusters, so the 95% interval is wide — read it as indicative, not a precise p-value. The effect holds whether the window is 4, 5 or 6 minutes long. The headline currently rests on 86 on-top hydration breaks, and the estimate-over-time chart on the main page shows whether it is stabilizing or fading.

06 — THE HEAT & ACCLIMATIZATION CHECK

The intuitive objection is that the swings are about heat, not the whistle — players from cool leagues wilting in the US summer. We tested it directly. For every team we built an *acclimatization gap*: the WBGT on match day minus the WBGT the squad is used to back home, mapping each player to his club's home city (391 clubs placed). If heat-displacement drove the drops, teams further from home should fall harder.

They do not. Across tournaments the biggest gap goes with the *smallest* drop, not the largest:

Tournament	heat gap	drop
Copa América 2024	+11°C	-10
Club World Cup 2025	+8°C	-23
World Cup 2026	+5°C	-19
Euro 2024	+4°C	-16

Within groups it is the same story. Among the Club World Cup clubs the gap→drop slope is +0.99 per °C [-0.30, +2.29] — if anything, clubs further from home dropped *less*. Pooled across national teams it is -0.12 per °C [-0.84, +0.55], essentially flat. So the big drops are not an acclimatization effect, and the high Club World Cup figure is structural — clubs regress more — not heat.

One honest caveat: the heat gap is tangled up with continent, league and fixture congestion, so this rules out the simple heat story rather than proving a precise null. And it never threatened the headline anyway — the same-teams control already holds a team's heat fixed.

07 — ALTITUDE

Altitude is a different stressor from heat — thin air, not thermoregulation — and only two venues are high (Mexico City and Guadalajara, both above 1,500 m). That is too few to test, and a cooling break is not built for it, so we note it and leave it alone.

08 — WHAT WE CAN AND CANNOT SAY

- **Small sample, for now.** Early in the tournament the intervals are wide; the estimate-over-time chart shows whether the effect stabilizes or fades.
- **Break durations, now measured.** ESPN's start/end-delay timestamps give an exact length for 86 of 86 on-top hydration breaks (median 2:51, 1:36–3:56); VAR and injury coverage is thinner. Whether *longer* breaks bite harder is inconclusive on the current sample — the slope's interval includes zero — so we don't lean on it.
- **WBGT is a shade estimate** from temperature and humidity; true on-pitch heat under sun runs higher.
- **The acclimatization gap is collinear** with continent, league and schedule, so it is suggestive, not a clean instrument.
- **No causal headline yet.** The agreed causal model — a two-way fixed-effects regression with match-clustered errors and the hydration×pre-momentum interaction — is held until the live sample is large enough for stable estimates.

09 — REPRODUCIBILITY

Every figure here and on the main page is computed deterministically from one file — the committed processed dataset — by the code in the repository. Nothing is hand-entered. The full source, the dataset, and this report are public.

github.com/valternunez/wc2026-momentum ↗